Available online at www.sciencedirect.com
ScienceDirect
Colloids and Surfaces A: Physicochem. Eng. Aspects 313–314 (2008) 608–611
COLLOIDS AND SURFACES A
www.elsevier.com/locate/colsurfa

Control of two-dimensional structure of tolanethioacetate self-assembled monolayers on Au(1 1 1)

Youngdo Jeong, Hoeil Chung, Jaegeun Noh *

Department of Chemistry, Hanyang University, 17 Haengdang-dong, Seongdong-gu, Seoul 133-791, Republic of Korea

Received 16 November 2006; accepted 20 April 2007
Available online 2 June 2007

Abstract

Nanometer-scale structure and binding condition of tolanethioacetate (TTA) self-assembled monolayers (SAMs) formed in various solutions on Au(1 1 1) were investigated by scanning tunneling microscopy (STM) and X-ray photoelectron spectroscopy (XPS). STM imaging revealed that two-dimensional (2D) ordered TTA SAMs with an overlayer structure of $(\sqrt{13} \times \sqrt{5})R24^\circ$ on Au(1 1 1) were formed in a polar protic solvent, while TTA SAMs with disordered phases were formed in a polar aprotic solvent and a nonpolar solvent. In addition, XPS measurements showed that the XPS spectra for TTA SAMs formed in a nonpolar solvent were markedly different from those formed in a polar protic solvent, indicating that the binding condition of TTA SAMs was influenced by the solvent properties. In this study, we found that 2D ordered TTA SAMs on Au(1 1 1) can be obtained by controlling the properties of solvents, in particular, by using a polar protic solvent.
© 2007 Elsevier B.V. All rights reserved.

Keywords: Self-assembled monolayers; Solvent effect; Tolanes; Domain structure; Scanning tunneling microscopy; X-ray photoelectron spectroscopy

1. Introduction

Organic self-assembled monolayers (SAMs) on metal surfaces have been extensively studied because of their technological applications in corrosion inhibition, nanopatterning, biointerfaces, molecular sensors, and nanodevices [1–5]. In particular, due to the interesting properties of π-conjugated aromatic SAMs, they are often used as a molecular template for both fundamental and application studies such as molecular electronics, molecular switching, and molecular transistors [6–9]. To understand self-assembly phenomena of various molecules and to control the two-dimensional structure of SAMs, several fundamental aspects of SAMs such as pH effects, annealing effects, temporal effects, and solvent effects have been extensively studied for further application [1,10]. Tolane compounds with a structure of a π-conjugated molecular rod are frequently used for molecular electronics because their conductivity can be tuned by changing the angle between two benzene rings through the introduction of functional groups to a benzene ring [11]. However, it was found that the thiol group can be easily oxidized to disulfides or other oxidized compounds during SAM formation in solution [12]. To avoid this problem, thioacetyl-terminated organic molecules showing a high chemical stability for oxidation in air or solution have been used for SAM formation [12–14]. After deprotection of the acetyl group in acidic or basic solution, the thiols or thiolates can easily form chemisorbed SAMs on a gold surface. Previous macroscopic studies have shown that the thioacetyl-terminated organic molecules could form chemisorbed SAMs under high concentration conditions (∼30 mM) or long immersion times (∼several days) in a pure solvent without any acid- or base-catalyzed cleavage of the thioacetate precursors [14,15].

It has been shown that solvent properties are major factors for controlling surface structures of SAMs [16,17]. High-quality carboxylic acid-terminated SAMs were obtained in a polar aprotic solvent compared with a polar protic solvent [16]. In this study, we report a method to obtain 2D ordered SAMs of thioacetyl-terminated tolanethioacetate (TTA) on Au(1 1 1) by controlling solvent properties under typical SAM preparation conditions (0.5 mM concentration and 1 day deposition), without any acid- or base-catalyzed reactions.

* Corresponding author. Tel.: +82 2 2220 0938; fax: +82 2 2299 0762.
E-mail address: jgnoh@hanyang.ac.kr (J. Noh).

0927-7757/$ – see front matter © 2007 Elsevier B.V. All rights reserved.
doi:10.1016/j.colsurfa.2007.04.132

2. Experiment

Tolanethioacetate (TTA) compound containing a π-conjugated phenyl group was synthesized by modifying a previously reported method [18]. Au(1 1 1) substrates were prepared by the vacuum evaporation of gold onto freshly cleaved mica sheets pre-baked at 330 °C for 2 h under a vacuum pressure of 10^{-7} to 10^{-8} Torr. The substrates were then annealed at 350 °C in the same vacuum chamber for 2 h to obtain an atomically flat, large, and single-crystal surface [19]. In order to examine the effect of solvent on the structure and binding condition of TTA SAMs on Au(1 1 1), TTA SAMs were prepared by dipping the Au(1 1 1) substrates into freshly prepared 0.5 mM ethanol, methanol (polar protic solvents), *N*,*N*′-dimethylformamide (DMF, a polar aprotic solvent), and toluene (a nonpolar solvent) solutions of TTA at room temperature for 1 day. The TTA SAM samples were then thoroughly washed with a pure solvent used to make the SAMs in order to remove physisorbed molecules on the SAM surfaces.

STM measurements were performed using a NanoScope E (Veeco, Santa Barbara, CA) with a commercial Pt/Ir (80:20) tip and all STM images were obtained in air using the constant current mode. A bias voltage (V_b) ranging from 200 to 700 mV and a tunneling current (I_t) ranging from 0.15 to 0.50 nA were applied between the tip and sample. X-ray photoelectron spectroscopy (XPS) spectra were obtained using a VG ES-CALAB 250 system with a monochromatic Al Kα X-ray source (1486.6 eV). The binding energies were calibrated by the Au(7/2) peak (83.9 eV) and the X-ray power was 200 W. The pass energy of the analyzer and the takeoff angle of photoelectrons were set at 40 eV and 90°, respectively.

3. Results and discussion

The STM images in Fig. 1 show surface morphologies of TTA SAMs on Au(1 1 1) formed in various solutions at room temperature after immersion for 1 day. The STM images of TTA SAMs in Fig. 1a and b show the formation of 2D ordered phases (region A) consisting of small sized domains and domain boundaries with wide disordered phases (region B) from ethanol and methanol. The structural features of the TTA SAMs were completely different from alkanethiol or aromatic thiol SAMs that contain the long-range ordered domains and sharp domain boundaries. It is reasonable to expect that the adsorption activity of acetyl group-protected sulfur atoms onto gold should be lower than that of sulfur atoms in alkane thiols or aromatic thiols because the acetyl group has an electron-withdrawing character. Therefore, TTA molecules can slowly adsorb onto gold during the initial stage of SAM growth. A previous kinetic study has shown that the growth kinetics of thioactyl-terminated oligo(phenylene ethyny-

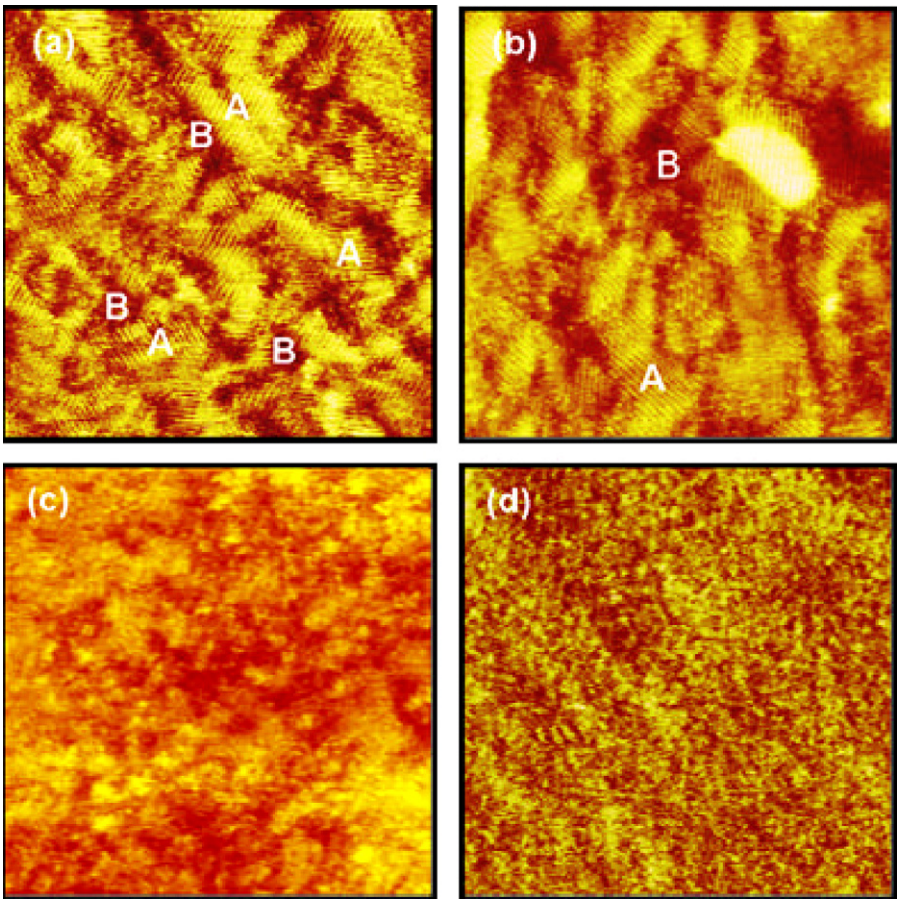

Fig. 1. STM images of TTA SAMs on Au(1 1 1) formed in 1 mM (a) ethanol, (b) methanol, (c) DMF, and (d) toluene solutions of TTA after immersion at room temperature for 1 day. All scan sizes were 60 nm × 60 nm.

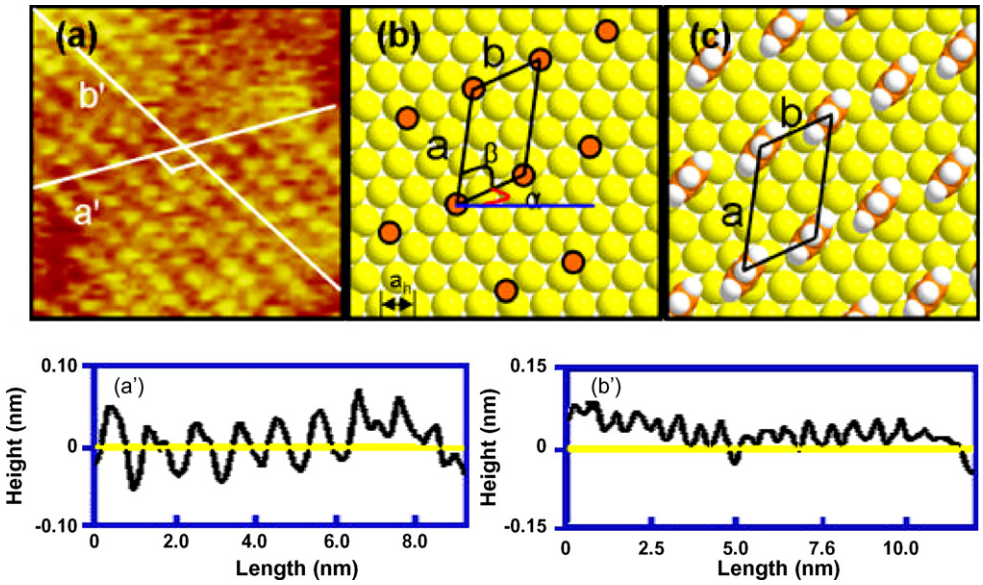

Fig. 2. (a) Molecularly resolved STM image of TTA SAMs on Au(1 1 1) obtained from region A of Fig. 1(b). The scan size was 9 nm × 9 nm. (a′) and (b′) Height profiles along lines a′ and b′ on the STM image. (b) Schematic structural model and (c) space-filled drawing model of the proposed structure for TTA SAMs.

lene)s (OPE) are much slower than that of alkanethiols, even in a base-catalyzed solution [12]. As a result, the slow adsorption step significantly affects the formation and growth of 2D SAMs. On the other hand, the STM images in Fig. 1c and d show the formation of disordered phases of TTA SAMs in DMF and toluene. From these results, we clearly revealed that the solvent choice is very important for TTA SAM fabrication with 2D ordered phase. Based on our STM results, it is assumed that the hydrogen atoms of polar protic solvents such as ethanol or methanol can be effectively transferred onto the oxygen atom of the acetyl group. As a result, the protected acetyl groups can be effectively deprotected on Au(1 1 1) surfaces during self-assembly and then spontaneously form SAMs. However, DMF and toluene are not capable to transfer hydrogen atoms onto the oxygen atom of the acetyl group because DMF and toluene are a polar aprotic solvent and a nonpolar solvent, respectively.

The molecularly resolved STM image in Fig. 2a shows a well-ordered structure of TTA SAMs on Au(1 1 1) formed after 1 day immersion in methanol. The cross-sectional profiles in Fig. 2(a′) and (b′) taken along lines a′ and b′, corresponding to the oblique unit cell in Fig. 2(a), show the periodic molecular arrangements of THT molecules on Au(1 1 1). From the high-resolution STM image, we extracted the lattice constants of the unit cell: $a = 10.3 \pm 0.2$Å $= \sqrt{13}a_h$, $b = 6.3 \pm 0.2$Å $= \sqrt{5}a_h$, $a = 24°$, and $b = 60°$ (see Fig. 2b), where $a_h = 2.88$ Å and represents the interatomic distance of the Au(1 1 1) lattice. The 2D packing structure of THT SAMs can be assigned as $a(\sqrt{13} \times \sqrt{5})$R24° structure. We suggest a schematic structural model and a space-fill drawing model of the proposed structure for TTA SAMs on Au(1 1 1), as shown in Fig. 2b and c. It was also found that all sulfur atoms occupy bridge sites of the Au(1 1 1) lattice. The average areal density for THT SAMs with the $(\sqrt{13} \times \sqrt{5})$R24° structure was calculated to be about 56.2 Å^2/molecule. On the other hand, OPE SAMs formed via acidic hydrolysis of thioacetate group or from the thiol have a (v3 × v3)R30° structure with a molecular density of 21.6 Å^2/molecule [13]. From these results, we demonstrated that the ordered THT SAMs with a low surface coverage can be obtained from ethanol or methanol.

In order to understand the solvent effect on the binding condition of TTA SAMs on gold, we observed XPS spectra in the S 2p region of TTA SAM samples prepared from two different solvent systems such as methanol (polar protic solvent) and toluene (nonpolar solvent). Fig. 3 shows large differences in the XPS S 2p spectra for the two different SAM samples, indicating that the solvent property markedly affects the adsorption condition of TTA molecules during the self-assembly processes. The S1 and S3 peaks in Fig. 3 were observed at 162.2 and 161.2 eV, respectively, and were assigned to the bound sulfurs. The S2 peak was observed at 161.2 eV and assigned to the unbound sulfurs. XPS spectra containing the S1 and S2 peaks were usually observed from the closely packed, well-ordered SAM samples prepared by alkanethiols or aromatic thiols [20–22]. It was found that the main difference in XPS spectra for the two SAM samples was the relative intensity of the S2 and S3 peaks. Although TTA SAMs containing the ordered phases formed in methanol had S3 peaks of weak intensity, other XPS peaks are nearly the same as those for well-ordered alkanethiol SAMs (Fig. 3a). On the other hand, TTA SAMs containing the disordered phases formed in toluene have relatively strong unbound sulfurs (S2 peak) and bound sulfurs (S3 peak) (Fig. 3b). Based on our XPS results, we suggest that the origin of the S3 peak is closely related to the adsorption configuration of TTA molecules. In addition, it is reasonable to assume that some of the sulfur atoms attached to the molecular backbone with a nearly lying-flat adsorption configuration in the disordered phase can interact strongly with the gold surface, resulting in the appearance of the S3 peak with a strong intensity. This assumption can be supported by a previous report that the S3 peak was often observed in low coverage alkanethiol SAMs containing the striped phase where the molecular

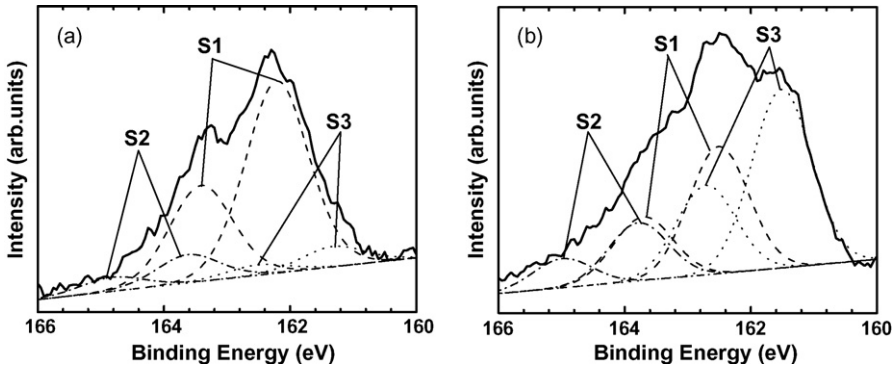

Fig. 3. XPS spectra in S 2p region of TTA SAMs on Au(1 1 1) formed in 1 mM (a) methanol and (b) toluene solutions after immersion at room temperature for 1 day.

backbone was oriented parallel to the surface [21]. From these results, we suggest that the binding condition of sulfur atoms can be changed according to a change in the adsorption configuration of molecules. In this study, we revealed that the binding condition of TTA molecules in SAMs largely depends on the 2D structures of SAMs according to whether the TTA SAMs are composed of the ordered phases or the disordered phase.

4. Conclusion

We have demonstrated that the surface structure and binding condition of thioacetyl-terminated TTA SAMs on Au(1 1 1) strongly depend on the solvent property. It was found that the 2D ordered TTA SAMs can be obtained by using polar protic solvents such as ethanol or methanol. The high-resolution STM study revealed that TTA SAMs formed in methanol on Au(1 1 1) have an overlayer structure of $(\sqrt{13} \times \sqrt{5})R24^\circ$. XPS measurements revealed that TTA SAMs containing the disordered phases formed in toluene show XPS spectra with a strong intensity at 161.2 eV, compared to TTA SAMs containing the ordered phases formed in methanol. Based on our XPS result, we suggest that the origin of the XPS peak observed at 161.2 eV is closely related to the adsorption configuration of TTA molecules.

Acknowledgment

This work was supported by the research fund of Hanyang University (HY-2006-I), Korea.

References

[1] J.C. Love, L.A. Estroff, J.K. Kriebel, R.G. Nuzzo, G.M. Whitesides, Chem. Rev. 105 (2005) 1103.
[2] T. Felgenhauer, C. Yan, W. Geyer, H.T. Rong, A. Gölzhäuser, M. Buck, Appl. Phys. Lett. 79 (2001) 3323.
[3] J.M. Tour, Acc. Chem. Res. 33 (2000) 791.
[4] H.D. Sikes, J.F. Smalley, S.P. Dudek, A.R. Cook, M.D. Newton, C.E.D. Chidsey, S.W. Feldberg, Science 291 (2001) 1519.
[5] J.F. Smalley, S.B. Sachs, C.E.D. Chidsey, S.P. Dudek, H.D. Sikes, S.E. Creager, C.J. Yu, S.W. Feldberg, M.D. Newton, J. Am. Soc. Chem. 126 (2004) 14620.
[6] A.H. Flood, J.F. Stoddart, D.W. Steuerman, J.R. Heath, Science 306 (2004) 2055.
[7] A. Nitzan, Annu. Rev. Phys. Chem. 52 (2001) 681.
[8] Z.J. Donhauser, B.A. Mantooth, K.F. Kelly, L.A. Bumm, J.D. Monnell, J.J. Stapleton, D.W. Price, A.M. Rawlett, D.L. Allara, J.M. Tour, P.S. Weiss, Science 292 (2001) 2303.
[9] C.R. Kagan, A. Afzali, R. Martek, L.M. Gignac, P.M. Solomon, A.G. Schrott, B. Ek, Nano Lett. 3 (2003) 119.
[10] Y. Jeong, C. Lee, E. Ito, M. Hara, J. Noh, Jpn. J. Appl. Phys. 45 (2006) 5906.
[11] J. Chen, M.A. Reed, Chem. Phys. 281 (2002) 127.
[12] L. Cai, Y. Yao, J. Yang, D.W. Price, J.M. Tour, Chem. Mater. 14 (2002) 2905.
[13] J.J. Stapleton, P. Harder, T.A. Daniel, M.D. Reinard, Y. Yao, D.W. Price, J.M. Tour, D.L. Allara, Langmuir 19 (2003) 8245.
[14] J.M. Tour, L. Jones II, D.L. Pearson, J.J.S. Lamba, T.P. Burgin, G.M. Whitesides, D.L. Allara, A.N. Parikh, S.V. Atre, J. Am. Chem. Soc. 117 (1995) 9529.
[15] Y. Kang, D. Won, S.R. Kim, K. Seo, H.-S. Choi, G. Lee, Z. Noh, T.S. Lee, C. Lee, Mater. Sci. Eng. C 24 (2004) 43.
[16] J. Noh, K. Konno, E. Ito, M. Hara, Jpn. J. Appl. Phys. 44 (2005) 1052.
[17] S.Y. Lee, J. Noh, E. Ito, H. Lee, M. Hara, Jpn. J. Appl. Phys. 42 (2003) 236.
[18] S.H. Huang, J.M. Tour, J. Org. Chem. 64 (1999) 8898.
[19] J. Noh, M. Hara, Langmuir 16 (2000) 2045.
[20] D.G. Castner, K. Hinds, D.W. Grainger, Langmuir 12 (1996) 5083.
[21] T. Ishida, N. Choi, W. Mizutani, H. Tokumoto, I. Kojima, H. Azehara, H. Hokari, U. Akiba, M. Fujihira, Langmuir 15 (1999) 6799.
[22] F. Buckel, F. Effenberger, C. Yan, A. Gölzhäuser, M. Grunze, Adv. Mater. 12 (2000) 901.